Name _____ _____ Baggett/Fall2026/207/Quiz7

first name last name

No Calculators.

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 1 for the reason.**

1. (**2** pts) If $f'(x) < 0$ for $1 < x < 6$, then f is decreasing on $(1, 6)$.

True **False**

Why? The Increasing/Decreasing Test: a negative derivative throughout an interval means the function falls across the whole of it. This follows from the Mean Value Theorem, not from checking points. /2

2. (**2** pts) If f and g are differentiable and $f(x) \geq g(x)$ for $a < x < b$, then $f'(x) \geq g'(x)$ for $a < x < b$.

True **False**

Why? **Being larger says nothing about growing faster. A big constant beats a rising line for a while.** /2

3. (**7** pts) Let $f(x) = x^3 - 3x^2 - 9x + 5$.

(a) Find the critical numbers of f, and classify each one as a local maximum, a local minimum, or neither. **(4 points)**

$f'(x) = 3(x-3)(x+1)$**, so the critical numbers are** $x = -1$ **and** $x = 3$**.** f' **goes** $+, -, +$**, so** $x = -1$ **is a local maximum and** $x = 3$ **is a local minimum.**

$$\begin{aligned} f'(x) &= 3x^2 - 6x - 9 && \text{Power Rule, term by term} \\ &= 3\left(x^2 - 2x - 3\right) = 3(x-3)(x+1) && \textbf{factor before anything else} \\ x &= -1,\ 3 && \text{a product is zero when a factor is} \end{aligned}$$

One test point in each piece: $f'(-2) = 3(-5)(-1) = 15 > 0$, $f'(0) = 3(-3)(1) = -9 < 0$, $f'(4) = 3(1)(5) = 15 > 0$.

So f increases, decreases, increases. f' turns $+$ to $-$ at $x = -1$ (**local max**, value $f(-1) = 10$) and $-$ to $+$ at $x = 3$ (**local min**, value $f(3) = -22$).

(b) On what interval(s) is f concave up and concave down? Give the inflection point as an ordered pair. **(3 points)**

$f''(x) = 6(x-1)$**, so** f **is concave down on** $(-\infty, 1)$ **and concave up on** $(1, \infty)$**. The concavity changes, so** $(1, -6)$ **is an inflection point.**

$f''(x) = 6x - 6 = 6(x-1)$, which is zero only at $x = 1$. It is negative for $x < 1$ and positive for $x > 1$, so f is **concave down on** $(-\infty, 1)$ and **concave up on** $(1, \infty)$. The sign *changes*, so there is an inflection point, and $f(1) = 1 - 3 - 9 + 5 = -6$ gives the point $(\mathbf{1}, \mathbf{-6})$. /7

4. (**6** pts) A function f is differentiable everywhere, and

$$f'(x) = (x+3)(x-1) \qquad \text{and} \qquad f''(x) = 2(x+1).$$

You are **not** given a formula for f.

(a) On what intervals is f increasing and decreasing, and what happens at each critical number? **(3 points)**

(b) On what intervals is f concave up and concave down, and where is the inflection point? **(3 points)**

Increasing on $(-\infty, -3)$ and $(1, \infty)$, decreasing on $(-3, 1)$; local max at $x = -3$, local min at $x = 1$. Concave down on $(-\infty, -1)$, concave up on $(-1, \infty)$; inflection point at $x = -1$.

(a) $f' = 0$ at $x = -3$ and $x = 1$. The parabola $(x + 3)(x - 1)$ opens upward, so it is positive outside $[-3, 1]$ and negative inside: f is **increasing on $(-\infty, -3)$ and $(1, \infty)$** and **decreasing on $(-3, 1)$**. So f' goes $+$ to $-$ at $x = -3$ (**local maximum**) and $-$ to $+$ at $x = 1$ (**local minimum**).

(b) $f'' = 0$ at $x = -1$, negative to the left and positive to the right: **concave down on $(-\infty, -1)$, concave up on $(-1, \infty)$**, and the bend changes, so $x = -1$ is an **inflection point**.

/6

A check that costs nothing: $f''(-3) = -4 < 0$, concave down at the local maximum, which is what a hilltop should be; $f''(1) = 4 > 0$, concave up at the local minimum. The two halves agree.

5. (3 pts) Suppose $f'(2) = 0$ and $f''(2) = 0$. What does the Second Derivative Test tell you about $x = 2$, and what should you do instead?

It tells you nothing – the test is inconclusive when $f''(c) = 0$. Fall back to the First Derivative Test: check the sign of f' on both sides of $x = 2$.

The Second Derivative Test only decides when $f''(c)$ is strictly positive or strictly negative. At $f''(2) = 0$ it is **silent** – all three outcomes are still possible, and $y = x^4$, $y = -x^4$ and $y = x^3$ at the origin are three functions with $f'(0) = f''(0) = 0$ and a minimum, a maximum and no extremum respectively.

/3

What to do instead: the **First Derivative Test** – look at the sign of f' just left and just right of $x = 2$. If it changes $+$ to $-$ it is a maximum, $-$ to $+$ a minimum, and if it does not change, neither.